

BlueVantage — Product Specification & Feature Map v1.0

Last updated: April 25, 2026 | **Stage:** Pre-MVP / Hackathon Build | **Market:** Netherlands North Sea Informed Fishery

1. PRODUCT VISION

BlueVantage is a fishing zone intelligence marketplace — combining environmental data, predictive species modeling, and a self-allocation bidding system to help Dutch commercial fishers maximize catch yield, minimize fuel waste, stay compliant with MPAs, and sustainably share the sea.

From: Guesswork-driven fishing trips
To: Data-informed, competitively allocated, sustainability-compliant zone selection

2. CORE PRODUCT PRINCIPLES

Principle	Implication
No crowding by design	Zones have hard capacity caps — recommendations are never broadcast universally
Fisher autonomy	Bidding is self-directed; the platform informs, not dictates
Data reciprocity	Fishers who share logbook data get more from the platform
Compliance built-in	MPA boundaries are non-negotiable hard blocks, not suggestions
Progressive complexity	MVP is simple map + zone cards; complexity layers in post-launch

3. USER PERSONAS

Primary: The Vessel Owner-Operator ("Jan")

- Operates 1–2 vessels, North Sea out of IJmuiden / Scheveningen / Den Helder
- Fishes 4–5 species seasonally (plaice, sole, herring, cod, mackerel)
- Pain: spends €800–1,200/trip on fuel; 30–40% of trips underperform
- Tech comfort: smartphone-native, uses WhatsApp for fleet comms
- Wants: confident zone choice before leaving port, fewer empty nets

Secondary: The Fleet Manager ("Roos")

- Manages 5–15 vessels for a cooperative or family business
- Needs aggregate view: which vessel goes where, total quota usage, compliance status
- Wants: dashboard, multi-vessel bid management, exportable reports

Tertiary: Regulator / Sustainability Partner (future)

- RWS, NVWA, fishing cooperatives
- Wants: MPA compliance data, stock pressure heatmaps, anonymized aggregate behavior

4. FEATURE MAP

Organized by Release Phase

● MVP — Hackathon Demo (Sprint: ~3 weeks)

M1 — Zone Intelligence Map

- Interactive North Sea map using **Mapbox GL JS** or **Deck.gl**
- Hexagonal H3 grid overlay (resolution 5 = $\sim 25\text{km}^2$ cells)
- Each hex cell color-coded by **Zone Score** (composite 0–100):
 - Sea surface temperature match for target species
 - Chlorophyll-a density (proxy for food chain activity)
 - Historical catch density (ICES data)
 - Current weather/wave viability
 - Distance from home port (fuel cost factor)
- MPA zones rendered as hard red-hatched exclusion areas (non-clickable)
- AIS vessel density shown as static layer (refresh every 15 min)

M2 — Zone Card

- Click any hex → side panel opens with:
 - **Zone Score** (0–100 with breakdown)

- **Predicted dominant species** (e.g., 62% Plaice, 28% Sole, 10% Other)
- **Estimated biomass window** (e.g., 400–700 kg per vessel-day)
- **Current allocation:** X of Y vessel slots filled
- **MPA status:** Clear / Warning / Blocked
- **Weather window:** wave height, wind speed for next 48h
- **Bid button** (if zone has open slots)

M3 — Species Preference Setup (Onboarding)

- Fisher selects their target species (multi-select from 5: Plaice, Sole, Herring, Cod, Mackerel)
- Gear type input (beam trawl, otter trawl, gillnet) — affects zone suitability scoring
- Home port selection (auto-calculates distance cost in zone scoring)
- This profile persists and filters the map scoring to that fisher's context

M4 — Self-Allocation Bidding (Core Mechanic)

- Each zone has a **capacity cap** (max concurrent vessels based on estimated stock pressure)
- Fisher clicks "Bid" → confirms zone + time window (morning / full day / overnight)
- **Bid currency for MVP:** simple first-come-first-served slot reservation (no financial bid yet)
- Confirmation shows: slot number, zone summary, estimated departure time window
- Zone slot count updates in real time once allocated
- Fisher can release a slot (cancellation) up to X hours before window

M5 — MPA Alert System

- All zone recommendations exclude MPAs by default (hard filter in scoring engine)
- If fisher manually navigates map toward MPA boundary: passive overlay warning
- Onboarding screen explains MPA system and fine exposure (€50k+)
- MVP: static MPA polygons from EMODnet / RWS Natura 2000 data

M6 — Logbook Data Incentive (Lightweight)

- Post-trip: fisher submits simple logbook entry (zone, species caught, weight, gear)
 - On submission: earns **priority bid credits** (e.g., +1 early-access slot per week)
 - Data feeds back into prediction model training pipeline
 - MVP: manual form submission; V2 automates via VMS/logbook API
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● V1.1 — Post-Hackathon (Month 1–3)

V1 — Live AIS-Driven Zone Downgrading

- Real-time AIS feed ingestion (MarineTraffic API or AISHub)
- As vessels move into a zone, live occupancy count updates
- Zone score dynamically penalized when approaching capacity
- Crowding pressure visible to all fishers browsing that zone

V2 — Financial Bidding System

- Introduce **zone credits** (earned via logbook sharing + subscription tier)
- Fishers spend credits to bid higher priority on contested zones
- High-demand zones go to highest bidder (within vessel capacity cap)
- Credit top-up available as in-app purchase (monetization layer)
- Ethical guardrail: credits are non-transferable, no secondary market

V3 — Quota Integration

- Fisher inputs their annual ITQ quota by species
- Platform tracks quota consumption per trip (based on logbook submissions)
- Zone recommendations weighted by remaining quota — no point sending fisher to plaice zone if plaice quota is exhausted
- Quota burn rate visualization: "At current pace, your sole quota runs out in 6 weeks"

V4 — Notification System (Tier-Based)

- **Standard:** Weekly zone forecast email / push notification
- **Pro:** 48h ahead push alerts when premium zones open up

- **Pro:** Real-time alert when a reserved slot in your shortlisted zone gets cancelled (take it)
 - **Fleet:** Aggregate fleet briefing — zone assignments per vessel for the week
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● V2.0 — Scale Phase (Month 4–12)

S1 — Fleet Dashboard

- Multi-vessel management view
- Zone assignment per vessel, real-time AIS tracking overlay
- Aggregate quota usage, earnings estimate, compliance status
- Exportable weekly reports (PDF/CSV) for cooperative management

S2 — Predictive Model Upgrade

- Upgrade from XGBoost tabular model → Spatial ML (Gaussian Process or GNN over H3 graph)
- Incorporate fisher logbook corpus (crowd-sourced ground truth)
- Species model expands beyond 5 to 10+ species
- Seasonal trend modeling: automatic model retraining per quarter

S3 — Sustainability Reporting Module

- Per-fisher carbon footprint estimate (fuel × distance × catch weight)
- Aggregate platform impact: tons of CO₂ saved vs. unoptimized routes
- MPA intrusion events avoided (platform-wide stat)
- Shareable sustainability certificate (for cooperatives / buyers who care about chain of custody)
- This is a **B2G revenue angle** — sell aggregate anonymized reports to RWS / EU fisheries bodies

S4 — Market Price Integration

- Connect to fish auction price feeds (Urker Visafslag, IJmuiden)
- Zone cards show **estimated trip revenue** not just yield (kg × current auction price)
- Alert: "Sole price spiked 18% today — high-value window to prioritize Zone 7B"

5. FEATURE PRIORITY MATRIX

Feature	Impact	Effort	Phase
H3 Zone Map + Scoring	★★★★★	High	MVP
Zone Card (species, yield, slots)	★★★★★	Medium	MVP
MPA Hard Exclusion Layer	★★★★☆	Low	MVP
Species Preference Onboarding	★★★★☆	Low	MVP
Slot-based Bidding	★★★★★	Medium	MVP
Logbook Incentive (manual)	★★★★☆	Low	MVP
Live AIS Zone Downgrading	★★★★★	High	V1.1
Credit-based Financial Bidding	★★★★☆	High	V1.1
Quota Integration	★★★★☆	Medium	V1.1
Tiered Push Notifications	★★★☆☆	Low	V1.1
Fleet Dashboard	★★★★☆	High	V2.0
Spatial ML Model	★★★★★	Very High	V2.0
Sustainability Reporting	★★★☆☆	Medium	V2.0
Market Price Integration	★★★★☆	Medium	V2.0

6. GEOSPATIAL DATA PIPELINE (Technical Reference)

[RAW DATA SOURCES]

CMEMS API → SST, Chlorophyll-a, Salinity, Currents (daily rasters)

KNMI / OpenMeteo → Wind, Wave height, Visibility (48h forecast)

AISHub / MT API → Vessel positions (15-min refresh)

ICES / WMR → Historical stock assessment (annual, spatial)

EMODnet / RWS → MPA polygons, Bathymetry, Substrate (static)

Fisher Logbooks → Catch location, species, weight (user-submitted)



[STAGE 1 — INGESTION & HARMONIZATION]

- Raster data snapped to H3 hex grid (resolution 5, ~25km²)
- Vector MPA polygons → binary mask per hex (0 = clear, 1 = blocked)
- AIS tracks → vessel-hours-per-hex aggregation
- PostGIS database for spatial queries
- Scheduled ETL: daily for environmental, 15-min for AIS



[STAGE 2 — PREDICTION ENGINE]

- Feature vector per hex cell:
[SST, chlorophyll, depth, substrate, season_week,
historical_catch_density, current_vessel_count,
wave_height, wind_speed, distance_from_port]
- Model: XGBoost multi-output classifier
 - Output: P(species_i | hex, time) for 5 species
 - Output: estimated_yield_range (kg/vessel-day)
- Zone Score = weighted sum of species match + yield + safety + distance penalty
- Model retrained monthly on new logbook submissions



[STAGE 3 — SERVING LAYER]

- REST API endpoint: GET /zones?lat=&lng=&radius=&species=&time=
- Returns: ranked hex list with zone scores, species probs, slot availability
- Cached at 15-min intervals for map tile performance



[STAGE 4 — FRONTEND VISUALIZATION]

- Mapbox GL JS + deck.gl H3HexagonLayer
- Color ramp: green (score 80–100) → yellow (50–79) → red (<50)
- MPA overlay: red hatched polygon layer (always on top)
- AIS layer: live vessel dots with MMSI tooltip
- Zone Card: triggered on hex click → fetches /zone/{hex_id}/detail

7. BIDDING SYSTEM LOGIC (MVP Rules)

Zone Capacity Formula:

$$\text{max_vessels(hex)} = \text{floor}(\text{estimated_biomass_kg} / \text{avg_vessel_catch_kg_per_day})$$

Capped at: max 8 vessels per hex per time window (North Sea scale)

Slot Reservation (MVP — first-come-first-served):

1. Fisher selects hex + time window (AM / Full Day / Overnight)
2. System checks: slots_available > 0 AND fisher quota not exhausted
3. If valid → slot reserved, zone occupancy count +1
4. Slot held for 2h; if fisher doesn't depart (no AIS movement), slot released
5. Released slots trigger push notification to waitlisted fishers

Credit Bidding (V1.1):

- Each fisher starts with baseline credits per subscription tier
- Additional credits earned via logbook submissions (1 entry = 5 credits)
- Contested zone auction: highest bidder wins slot (blind auction, 24h before window)
- Minimum bid = 10 credits; Standard tier cap = 50 credits per zone per window
- Pro/Fleet tiers have higher credit caps → monetization lever

8. SUBSCRIPTION TIER ARCHITECTURE

Tier	Price	Zone Access	Species	Bidding	Notifications	Logbook
Free	€0	Read-only map, no bidding	1 species filter	None	Weekly email	Manual entry, no reward
Standard	€150/month	5 slot reservations/week	5 species	First-come slots	Weekly push	+5 credits per entry
Pro	€250/month	15 slot reservations/week	5 species + custom	Credit bidding	Real-time push + 48h alerts	+10 credits per entry
Fleet	€600/month	Unlimited (multi-vessel)	All species	Priority credits	Fleet briefing dashboard	API logbook sync

9. SUCCESS METRICS (KPIs by Phase)

MVP / Hackathon

- Demo: 5 simulated zone scores rendered correctly on map
- Demo: Bidding flow end-to-end in under 60 seconds
- Demo: MPA exclusion visibly working on map
- Judge question: can articulate crowding solution clearly

3 Months Post-Launch

- 20 paying vessels (Standard+)
- Logbook submission rate >60% of active users monthly
- Zone score accuracy: predicted vs. actual catch within $\pm 25\%$ (logbook validation)
- 0 MPA violations by platform users

12 Months

- 150+ paying vessels

- €270k ARR
- Average fisher trip yield improvement: +12% vs. baseline
- Logbook database: >500 trip entries (sufficient for model retraining)

10. OPEN QUESTIONS / RISKS TO RESOLVE

Question	Owner	Priority
Do Dutch fisher cooperatives (e.g., VisNed) allow data sharing agreements?	Business Dev	High
ICES data granularity — is it per hex or regional aggregate only?	Data Engineer	High
VMS (Vessel Monitoring System) data access — government-held, needs NVWA partnership	Business Dev	Medium
Logbook legal status — can we accept self-reported logbook entries or must they be official?	Legal	Medium
Credit bidding fairness — does it disadvantage small operators vs. fleet players?	Product	Medium
AIS refresh rate — does 15-min refresh create arbitrage window for gaming the system?	ML Engineer	Low
